

Pre-Feasibility Study for Establishing Poultry Feed Factory in Irbid



2017



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1. Executive summary

Irbid is located 71 km north of the capital Amman, and about 20 km south of the Syrian borders. It is the second largest city in Jordan after Amman. The city and its surroundings equals to 650 thousand. The estimated area of the city is about 30 square kilometers.

Irbid has Fertile plains, and many valleys. It also has many poultry farms, but lacks fodder factories with high quality and moderate prices, hence the idea of this project is to establish a poultry feed factory in Irbid governorate.

The idea of the project is to establish a factory to produce poultry feed in Irbid governorate to support livestock — poultry. Poultry is considered one of the important projects in the field of food security. It and depend on agricultural products. Therefore, this project could be at the level of small and medium enterprises, especially in Irbid due to the large poultry farms where the governorate has 442 poultry farms across the different districts. The proposed product is fodder for various types of poultry such as broiler, hens and chickens.

The project aims to manufacture poultry feed using food waste that can be obtained from farms, mills and food oils factories. There will be an agreement with these parties to collect food waste to be recycled into poultry feed. The justification for this project is to provide various types of poultry feed in addition to revitalize the industrial sector in Irbid governorate

Table1 : Study major results

Indicator	
Total Investment Requirement (JD)	782,910
Internal Rate of Return (IRR)	%31.98
Pay-Back Period (year)	5
Net present value (JD)	2,388,672
Job Opportunities provided by the project	16
Annual Revenue Projections (10 years)	17,239,628
Total Operating Expenses (10 years)	13,554,053
Net Profit (10 years)	2,163,332

2. Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Million distributed among 12 Governorates.

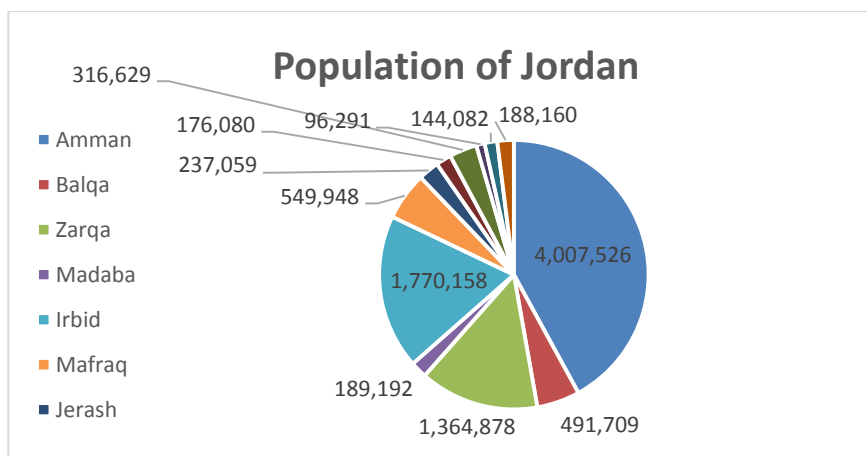
Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1: Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafraq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table2 : Population Distribution according to Age Group:

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	%18.2
15-64 Years	%61.02	%30.7	%28.6
More than 65 Years	%3.94	%1.6	%1.6

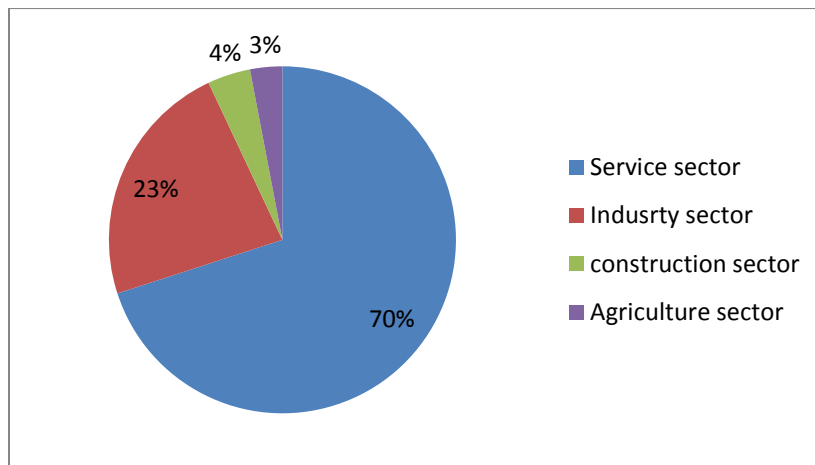
Department of Statistics (DOS), 2015.

Overview of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure2 : Sectors Contribution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US\$3.8 trillion market and comprising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and medium sized enterprises for 2015-2019. This shows the commitment of the Jordan

government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.

Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments,

the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance. The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

furthermore, in it endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency

permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Investment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

Investment Climate in Irbid:

Under the provisions of fifth article of the investment law No. 30 of 2014 and Income tax regulations in the less developed areas, the reduction of the income tax system on the investment projects are clarified in the table below:

Table3 : Income Tax and Exemptions

Within the Development Zone or Free Zone	Less Developed Areas
Income Tax is 5%	Income Tax is 20%

As per Investment Law categorization, Irbid is classified in Class (C) where investors enjoys 60% exemption on Income tax for 20 years, where net income tax after reduction should not be less than 5%.

3. Market analysis

Project description

The idea of this project is to establish a factory to produce poultry feed, where the project will target the owners of poultry farms in Irbid governorate in particular and the rest of the governorates of the Kingdom in general in addition to the Ministry of Agriculture. The proposed project will manufacture poultry feed from food waste that can be obtained from farms, mills and edible oils factories.

Project objectives

The project's objectives are:

- Supplying poultry farms with different types of feed.
- Revitalizing the industrial sector in Irbid.
- Providing job opportunities for the governorate's residents
- Achieving profit return to the project owners

Proposed project

The proposed project will produce different types of feed used for different types of poultry such as:

- Broiler
- Laying hens
- Chicken

The feed consists mainly of food waste that consists of soybeans, corn and olives from the farms, mills and edible oil factories as they will be contracted to buy food waste.

Target segments

- Owners of poultry farms (Irbid, in particular, and the Kingdom in general)
- The Ministry of Agriculture, which usually issues several bids to buy feed to support some small projects

The following table shows the number of poultry farms in Irbid by districts:

Table 4: Number of poultry farms according to district

District	Number of poultry farms
Qasabat Irbid	91
Ramtha	86
Al- kora	76
Bani kenanah	60
Bani ebead	55
Al- Taiba	38
Northern Almazar	20
Al-Wasatya	14
Northern Agwar	2

Source: Irbid information bank

Market size analysis

The following table shows estimates for poultry numbers in Jordan during the period (2013-2016) based on previous studies conducted by the Directorate of livestock

Table5 : Number of poultry during the years (2013-2016)

Chicken type	2013	2014	2015	2016
Laying hens	8,360,000	6,412,274	5,390,000	7,480,000
Broiler	29,730,800	30,800,000	30,093,800	30,948,500
Chicken	4,102,340	2,914,780	4,289,230	4,180,000

The following table shows the average annual consumption of feed for each type of poultry, taking into account the life cycle of broiler is 40 days:

Table6 : The annual consumption rate during the years (2013-2016)

The amount of feed poultry consumption in tons	2013	2014	2015	2016
Lying hens	457,710	351,071	295,102	409,530
Broiler	148,654	154,000	150,469	155,544
Chicken	187,169	132,986	195,696	192,712
Total (ton)	793,533	638,056	641,267	757,786

Based on the data (as feed consumption in the kingdom is 757,786 tons and sales prices for fodder is approximately JD 325 per ton), the total size of the Jordanian market was calculated and estimated. The following table shows the market total volume of feed during 2017:

Table 7: Total Jordanian market size

productive capacity analysis	
Size of feed consumption per ton (2016)	757,786
Selling rate feed (JD/ton)	325
Total market size/JD	246,280,450
Total	246,280,450

Competitors analysis

According to information from the Department of industry and Commerce and chambers of industry both in Amman and Irbid, there are number of competitors including: Al- Takmelia for feed additives, Al-Nukhba for agricultural fertilizer and veterinary medicines industry, Jordan feeds, Al-Qazaf factory for feeds, Nmu for Local templates and feed additives, Abd Al-Qader Mohammad Othman, Al-Takamol for Fertilizers and pesticides, Al-Shafaq industry, Fodder, Fakhre Mohammad Hamed Khrashfeh, International Matahen nameesh company, Omar Al-malah company, Sonqrot, United for the pharmaceutical industry, Al-jood for poultry feed,

Mahmoud Makahleh, Provimi Jordan feed concentrates industry, Australian-American trade and investment, Jarosha Mohammad Al-msaa'feh for feeds.

Price analysis and pricing policy

After studying and analysing the market, the selling price is determined at JD 325 per ton of poultry feed.

Expected Market share

Table 8: Expected Market share

Year	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
Expected market share	0.25%	0.35%	0.45%	0.55%	0.65%	0.75%	0.85%	0.95%	1.05%	1.15%

Expected revenues

Growth rate of poultry feed consumption per year is equal to 2%, which means that consumption of all types of poultry feed during 2017 is 757,786 tons.

Since the market price per ton of poultry fodder in 2017 equals approximately JD 325, then the size of the market reached JD 246,280,450 /year. Sale price per ton is fixed over the next 10 years because prices in recent years fluctuates from high to low. Therefore, revenues would be as follows:

Table 9: Expected revenues for 10 years

year	2017	2018	2019	2020	2021
Revenues	615701	861981	1108262	1354542	1600823

Year	2022	2023	2024	2025	2026
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Revenues	1847103	2093383	2339664	2585944	2832225
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4. Technical analysis

Required human resources

Table 10: Required human resources

Job title	No.
Indirect employees	
General manager	1
Accountant	1
Marketing officer	1
Guard	2
Maintenance technician	1
Total indirect employees	6
Direct employees	
Agricultural engineer	1
Driver	1
Manufacturing employee	8
Total direct employees	10
Grand total	16

The following shows the description for all jobs:

- General Manager: propose policies for the poultry feed factory, prepare and follow up private plans, technical and administrative supervision of staff, development of procedures and ensuring occupational health and safety requirements.

- Accountant: follow up the financial issues and audit.
- Marketing Officer: study the market needs, pursue publicity activities, assess their effectiveness, control and organize sales and marketing products.
- Guard: guarding the factory, make sure only accessed employees can enter the factory, and protect the factory from any robbery, especially on night shift.
- Maintenance technician: responsible for actions and activities of electric and mechanical maintenance, follow up the implementation of plans and programs established to ensure continuity of production process and quality at full capacity and lowest possible cost, and provide the necessary spare parts, and technical reports.
- Agricultural engineer: examine and advise on the application of science and engineering technology in agricultural production and natural resource management to resolve problems such as agricultural production, and the environmental impacts of agriculture and post-harvest processing agricultural products. In addition to planning, supervision, management of agricultural projects, construction of irrigation and drainage canals and water control.
- Driver: driving company transporting vehicles to transport feed to distribution outlets and consumers.

Location analysis

The factory will be established on an area of 2 donum including production area, storage, Offices area and outdoor space, in addition to extra free space in case of expanding the project in the future.

Production process description

Manufacturing procedure

The manufacturing procedure starts by receiving, and unloading and preparing raw materials, then crushing, pressing, and cooling it. Finally, feed is ready to be filled and packed to become ready for sale.

Required raw material

Required Raw materials are (soya, corn, and olives) which are taken from the remnants of vegetable oils factories, and (Hay) from farms, and (bran) from mills' remnants, and other feed additives consisting mainly of vitamins and flavouring.

Price per ton of farm, factories and Mills waste is: JD 170/ton

price per ton of other additives: JD 80 /ton

Technical requirements analysis

Land and construction works

The project will be established on a 2000m² land The following table shows the required land

Table 11: The required land for the project

Land	Area (m2)
Required land	2,000

The requires construction works as follows:

Table 12: Required construction works

Construction work	Area (m2)
Building work	1,000
Building design and schemes	2,000

Excavation work

1,000

Machinery and equipment

The following table shows the required Machinery and equipment for the project

Table 13: Required Machinery

Machinery and equipment	Quantity
Testing equipment	1
Fluid collection tanks	3
Blenders	3
Electronic balance	1
Filter presses	2
Crusher/grinder	2
Cooler	1
Crusher	1
Vibrators	2
Forklift	1
Packing equipment	1
Generator	1
Others	1

Furniture Requirements

The following table shows the required Furniture and furnishings in the project:

Table 14: Required Furniture and furnishings

Furniture and furnishings	Quantity
---------------------------	----------

Legal procedures

The steps needed to obtain licenses and approvals for project establishment:

Ministry of Industry and Trade is contacted to open a trade and commercial name record, and therefore the company registration certificate, lease, should be submitted to register a limited liability company in Irbid governorate through the Directorate of Industry and Trade. An identification card of the applicant for the professions licence must also be submitted. In addition, The following documents must be submitted

- Enrolment policy at the Chamber of Industry and Commerce.
- A bank deposit letter.
- Occupancy permit: which requires the project site map and the approval of the municipality

The municipality is contacted for approval and the following fees are paid:

- Ratio of capital (decided by the municipality) X profession fees
- JD 15 Declaration fees
- waste fee: 20% of the fees for the profession
- The health fee

Ministry of Environment: approval requires providing the following documents:

- land coordinates.
- Registration certificate (industry and Commerce)
- Territorial map
- Zoning scheme

Note that the project should be established in an industrial area and documents are sent to the environmental Department in Irbid, and then transferred to the Department of environment and then submitting it to the department of environment and mobilize technical model, then posting documents on Licensing Committee.

Irbid Chamber of industry: registration at the Chamber of industry via commercial registration, lease and the identity of the person, And the fee is according to the project's capital, where it equals to 600 JD if the project's capital ranges from (500,000 – 999,999) JD.

5. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is 11.61% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Income tax on the net project income is 5 % during the life time of the project.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Assumptions

Table 15: General Assumptions

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table 16: Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table 17: Work Time

Work Time	
Weeks/year	52
Working days/week	7
Working hours/day	8

Table 18: Annual Growth Rates Assumptions

Annual Growth Rates	Average
---------------------	---------

Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	3.00%
Annual Expenses Increase Rate	3.00%
Annual Increase Rate in Employees Salaries	4.1%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table 19: Expenses Assumptions

Expenses Assumptions	
Raw Materials and Packaging from Total Revenues	60.00%
Utilities Expenses from Total Revenues	5.0%
Maintenance Expenses from Total Revenues	1.50%
Depreciation Expenses from Total Revenues	1.00%
Insurance Expenses from Total Assets Value	0.75%
Marketing Expenses from Total Revenues	0.50%

Table 20: Income Tax Assumptions

Income Tax Assumptions	Value
(1) Average Income Tax in Jordan	5%
(2) Income Tax Deduction	10%

جدول 1 : فرضيات الميزانية العمومية

النسبة	فرضيات الميزانية العمومية
%100	النقد المحصل خلال الفترة من الإيرادات المستحقة
%0	النقد المحصل خلال الفترة اللاحقة
%83	النقد المدفوع خلال الفترة من المصاريف التشغيلية خلال السنة المالية
%17	النقد المدفوع خلال الفترة اللاحقة
%8	مخزون نهاية المدة

Other Assumptions	Value
(1) Annual Monthly Salaries have been calculated after multiplying monthly salaries by:	16

جدول 2 : علاوة المخاطرة

Risk Premuim	
(1) Risk Free Rate of Return	6.50%
(2) Return to Debt Maturity	9.48%
(3) Market Risk Premuim	10.12%
Income Tax Rate	14.00%

(4) Beta	0.89
Growth Rate	4.00%
(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report	(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report
(2) Source :Central Bank of Jordan	(2) Source :Central Bank of Jordan
(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report	(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report
(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu	(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table 21: Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%9.48
Market Risk Premium	%10.12
Income Tax Rate	%5

Bets	0.89
Equity	313,164
Loans Value	469,746
Weighted Average Cost of Capital	%11.61

Capital Expenditures

The estimated cost of the project is 782,910 JD and it covers land costs, construction works, machinery and equipment, furniture, pre-operating expenses and working capital. The following tables summarize the details of these expenses:

Table 22: Land and Construction work expenses

Expenses	Price (JD/m2)	Area (m2)	Total Price (JD)
Required Land	15	2,000	30,000
Construction Works	275	1000	275,000
Design and blueprints	7	2000	14,000
Excavation	5	1000	5,000
Soil Test	-	-	200
Other Works			14,710
Total			338,910

Table 23: Machinery capital expenses

Machinery and equipments	No.	Price (JD/unit)	Total (JD)
Testing equipment	1	7,000	7,000
Fluid collection tanks	3	500	1,500
Blenders	3	7,000	21,000
Electronic balance	1	2,000	2,000

Filter presses	2	16,880	33,760
Crusher/grinder	2	6,560	13,120
Cooler	1	5,005	5,005
crusher	1	4,793	4,793
Vibrators	2	4,934	9,868
Forklift	1	31,300	31,300
Packing equipment	1	10,798	10,798
Generator	1	23,000	23,000
Others	–	-	1,000
Total			164,144

Table 24: Furniture Capital Expenses

Description	No.	Price (JD/unit)	Total (JD)
Furniture	-	-	7,000
Total			7,000

Table 25: Transportation and Vehicles Capex

Transportation and Vehicles	No	Unit Cost/JOD	Total Cost/JOD
Pick up	2	23,400	46,800
Total			46,800

Table 26: Pre Operating Expenses

Pre-operating expenses	Estimated cost (JD)
Governmental fees	3,000
Transportation	750
Marketing	2,500
Training	2,500
Total	8,750

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table 27: Working Capital

Working Capital	No. of Months	%	Estimated Cost (JD)
Operating expenses	3	%25	129,909
General and administrative expenses	3	%25	2,253
Marketing expenses	3	%25	770
Indirect Salaries	3	%25	13,200
Total			146,132

Table 28: Expenses Summary

Capital expenditure summary	Cost (JD)	Contingency	Cost (JD)	%
Land	30,000	10%	33,000	%4.2
Construction works	308,910	10%	339,801	%43.4
Machinery & Equipment	164,144	10%	180,558	%23.1

Furniture	7,000	10%	7,700	%1.0
Vehicles	46,800	10%	51,480	%6.6
Pre-Operating Expenses	8,750	10%	9,625	%1.2
Working capital	146,132	10%	160,746	%20.5
Total (JD)	711,736		782,910	100%

Sources of Funding

The financing structure of the project consists of loans and Equity. The investment cost of this project is estimated at (782,910) JD, where 60% of the project will be financed through loans (469,746) JD, and the remaining (40%) through (Equity) with (313,164) JD. The following table summarizes the general structure of the required financing:

Table 29: Sources of Fundings

Funding Sources	Amount	Percentage
Equity	313,164	%40
(Self Financing)	469,746	%60
Loans	782,910	%100

Total	المبلغ	النسبة
Use of Fund:	612,539	%78.24
Capital Expenditures	9,625	%1.23
Pre-Operating Expenses	160,746	%20.53
Working Capital	782,910	%100.00

Operating Expenses

The following table summarizes the expected operating expenses for the proposed project (2018-2027):

Table 30: Operating Expenses

Description	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
Raw Materials and Packaging from Revenues	369,421	517,189	664,957	812,725	960,494	1,108,262	1,256,030	1,403,798	1,551,567	1,699,335
Utilities from Revenues	30,785	43,099	55,413	67,727	80,041	92,355	104,669	116,983	129,297	141,611
Maintenance from Revenues	9,236	12,930	16,624	20,318	24,012	27,707	31,401	35,095	38,789	42,483

Description	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
Disposables from Revenues	6,157	8,620	11,083	13,545	16,008	18,471	20,934	23,397	25,859	28,322
Direct Salaries	56,800	59,129	61,553	64,077	66,704	69,439	72,286	75,249	78,335	81,546
Others	47,240	64,097	80,963	97,839	114,726	131,623	148,532	165,452	182,385	199,330
Total	519,63	705,06	890,59	1,076,23	1,261,98	1,447,85	1,633,85	1,819,97	2,006,23	2,192,62
	8	3	3	2	5	7	2	5	2	8

General and Administrative Expenses

The following table summarizes the expected general and administrative expenses for the proposed project (2018-2027):

Table 31: General and Administrative Expenses

Description	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
Telecom	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Hospitality	250	258	265	273	281	290	299	307	317	326
Stationary and Printing	250	258	265	273	281	290	299	307	317	326
Transportation	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566

Training	700	721	743	765	788	811	836	861	887	913
Insurance	4,594	4,732	4,874	5,020	5,171	5,326	5,486	5,650	5,820	5,994
Miscellaneous	819	844	869	895	922	950	978	1,008	1,038	1,069
Total	9,013	9,284	9,562	9,849	10,145	10,449	10,763	11,085	11,418	11,761

Marketing Expenses

Table 32: Marketing Expenses

Description	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
Marketing Expenses	3,079	4,310	5,541	6,773	8,004	9,236	10,467	11,698	12,930	14,161

Human Resources

The total annual salaries and monthly wages has been estimated on 16 months for the purpose of considering annual increments on salaries as well as payments related to social security and health insurance,... etc.

The following table summarizes the expected annual salaries and wages, considering annual increment on salaries as per the income statement:

Table 33: Human Resources

Job Title	No.	Monthly Salary	Total Monthly Salary	Total Annual Salaries
Indirect employees				
General manager	1	1,500	1,500	24,000
Accountant	1	500	500	8,000
Marketing officer	1	500	500	8,000
Guard	2	200	400	6,400
Maintenance technician	1	400	400	6,400
Total indirect employees	6		3,300	52,800
Direct employees				
Agricultural engineer	1	800	800	12,800
Driver	1	350	350	5,600
Manufacturing employee	8	300	2,400	38,400
Total direct employees	10		3,550	56,800
Grand total	16		3,550	109,600

Table 34: Expected Annual salaries

Job Title	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
Indirect employees										
General manager	24,000	24,984	26,008	27,075	28,185	29,340	30,543	31,796	33,099	34,456

Job Title	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
Accountant	8,000	8,328	8,669	9,025	9,395	9,780	10,181	10,599	11,033	11,485
Marketing officer	8,000	8,328	8,669	9,025	9,395	9,780	10,181	10,599	11,033	11,485
Guard	6,400	6,662	6,936	7,220	7,516	7,824	8,145	8,479	8,826	9,188
Maintenance technician	6,400	6,662	6,936	7,220	7,516	7,824	8,145	8,479	8,826	9,188
Total indirect employees	52,800	54,965	57,218	59,564	62,006	64,549	67,195	69,950	72,818	75,804
Direct employees										
Agricultural engineer	12,800	13,325	13,871	14,440	15,032	15,648	16,290	16,958	17,653	18,377
Driver	5,600	5,830	6,069	6,317	6,576	6,846	7,127	7,419	7,723	8,040
Manufacturing employee	38,400	39,974	41,613	43,319	45,096	46,945	48,869	50,873	52,959	55,130
Total direct employees	56,800	59,129	61,553	64,077	66,704	69,439	72,286	75,249	78,335	81,546
Grand total	109,600	114,094	118,771	123,641	128,710	133,987	139,481	145,200	151,153	157,350

• Depreciations

The following tables illustrate cost of Capex and Depreciation rates:

Table 35: Capex and Depreciation Expenses

Capex Cost and Annual Depreciation Rates	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Land	33,000	%0.0	%0.0
Construction Works	339,801	%5.0	%0.0
Machineries	180,558	%10.0	%2.0

Furniture	7,700	%10.0	%5.0
Vehicles	51,480	%10.0	%0.0

Table 36: Depreciations and Additions on Construction Works

Capex, Annual Additions and Depreciations	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
Total Fixed Assets	612,539	616,536	620,623	624,805	629,083	633,459	637,938	642,521	647,210	652,010
Total Depreciation Expenses	40,964	41,364	41,772	42,190	42,618	43,056	43,504	43,962	44,431	44,911
Total Accumulated Depreciation	40,964	82,327	124,100	166,290	208,908	251,964	295,468	339,430	383,861	428,772
Total Additions	0	3,996	4,088	4,182	4,278	4,377	4,478	4,583	4,690	4,800
Total Net Book Values	571,576	534,208	496,524	458,515	420,174	381,495	342,470	303,091	263,349	223,238

Loan

The table below summarizes all details related to the loan such as annual installments and payment methods for the remaining amount of the loan for each year of the project.

Table 37: Loan Details

Loan Value	469,746
Annual Interest Rate	9.48%
Loans Period	5
Grace Period	2017
Loan Starts at year:	122,276
Annual Payment	5

Year	Annual Payment	Interest	Capital	Loan Remaining Value
2017				469,746
2018	122,276	44,532	77,744	392,002
2019	122,276	37,162	85,114	306,888

2020	122,276	29,093	93,183	213,705
2021	122,276	20,259	102,017	111,688
2022	122,276	10,588	111,688	0

Income Statement

من خلال جميع المعلومات والتحليلات المالية للمصاريف المالية للمشروع تكون قائمة الدخل التقديرية لمسنوات من (2026-2017) كما هي موضحة في الجدول التالي:

Table 38: Income Statement

Description	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
Revenues										
Sales Revenues	615,70	861,98	1,108,2	1,354,5	1,600,8	1,847,1	2,093,3	2,339,6	2,585,9	2,832,2
Gross Operating Revenues	615,70	861,98	1,108,2	1,354,5	1,600,8	1,847,1	2,093,3	2,339,6	2,585,9	2,832,2
Operating Expenses	519,63	705,06	890,59	1,076,2	1,261,9	1,447,8	1,633,8	1,819,9	2,006,2	2,192,6
Gross Operating Profit	96,063	156,91	217,66	278,310	338,838	399,246	459,532	519,689	579,712	639,597
Gross Profit Percentage	%16	%18	%20	%21	%21	%22	%22	%22	%22	%23
Salaries and Benefits (Indirect	52,800	54,965	-57,218	-59,564	-62,006	-64,549	-67,195	-69,950	-72,818	-75,804
General and Administraive	-9,013	-9,284	-9,562	-9,849	-10,145	-10,449	-10,763	-11,085	-11,418	-11,761
Markeing Expenses	-3,079	-4,310	-5,541	-6,773	-8,004	-9,236	-10,467	-11,698	-12,930	-14,161

Description	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
Pre Operating Expenses	-9,625									
Gross Indirect Expenses	74,517	68,559	-72,322	-76,186	-80,155	-84,233	-88,425	-92,734	-97,166	101,725
Income before Interest, Depreciation and Tax	21,546	88,360	145,347	202,124	258,682	315,013	371,107	426,955	482,547	537,871
Fixed Assets Depreciations	-	-	41,772-	42,190-	42,618-	43,056-	43,504-	43,962-	44,431-	44,911-
	40,964	41,364								
Income before Tax and Interests	-	46,996	103,575	159,934	216,064	271,957	327,604	382,993	438,116	492,960
Bank Interests	-	-	29,093-	20,259-	10,588-	0-	0-	0-	0-	0-
	44,532	37,162								
Income Before Tax	-	9,835	74,482	139,674	205,476	271,957	327,604	382,993	438,116	492,960
	63,950									
Income Tax	0	0	719-	1,746-	2,568-	3,399-	4,095-	4,787-	5,476-	6,162-
Net Profit	63,950	9,835	71,605	132,691	195,202	258,359	311,223	363,843	416,210	468,312
Net Profit Percentage	%10-	%1	%7	%10	%13	%15	%15	%16	%17	%17
Compulsory Reserves	0	983-	7,376-	13,793-	20,291-	26,856-	32,351-	37,821-	43,264-	48,680-
Retained Earnings	63,950	55,099	9,346	128,768	304,450	536,973	817,074	1,144,5	1,519,1	1,940,6

Expected Cash flows Statement

The table below summarizes project cash flows statement, where net cash flows are positive during all years as below:

Table 39: Expected Cashflows

	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
Cash Inflows from Operating Activities										
Net Profit	-63,950	9,835	71,605	132,69	195,20	258,35	311,22	363,843	416,210	468,312
Bank Interests	44,532	37,162	29,093	20,259	10,588	0	0	0	0	0
Depreciation	40,964	41,364	41,772	42,190	42,618	43,056	43,504	43,962	44,431	44,911
Total Operating Cashflows before	21,546	88,360	144,62	200,37	256,11	311,61	367,01	422,168	477,070	531,709
Additions to Working Capital			8	8	4	4	2			
Inventory (Increase/Decrease)	30,785-	12,314-	12,314-	12,314-	12,314-	-	-	12,314-	12,314-	12,314-

	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
						12,314	12,314			
Accounts Receivables (Increase/Decrease)	0	0	0	0	0	0	0	0	0	0
Accounts Payables (Increase/Decrease)	25,036	6,276	6,294	6,312	6,331	6,351	6,371	6,393	6,415	6,438
Working Capital (Increase/Decrease)	5,749-	6,038-	6,020-	6,002-	5,983-	5,963-	5,943-	5,922-	5,899-	5,876-
Net Cash flows from Operating Activities	15,797	82,322	136,450	189,138	242,425	295,452	348,784	401,884	454,742	507,347
Cash flows from Investments Activities										
Fixed Assets (Procurement)	-	3,996-	4,088-	4,182-	4,278-	4,377-	4,478-	4,583-	4,690-	4,800-
	612,539									
Net Cashflows from Investments	-	3,996-	4,088-	4,182-	4,278-	4,377-	4,478-	4,583-	4,690-	4,800-
	612,539									

	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
Activities	9									
Cashflows from Financing Activities										
Capital	313,16									
	4									
Loan Amortization	77,744-	85,114-	93,183-	-	-	0	0	0	0	0
				102,01	111,68					
				7	8					
Bank Interest Rate	44,532-	37,162-	29,093-	20,259-	10,588-	0-	0-	0-	0-	0-
Loans	469,74	0	0	0	0	0	0	0	0	0
	6									
Net Cash flows from Financing	660,63	122,27	122,27	122,27	122,27	0	0	0	0	0
Net (Increase/Decrease) in Cash	63,892	-43,950	10,087	62,681	115,87	291,07	344,30	397,301	450,052	502,547
Cash flows at the Beginning of	0	63,892	19,942	30,028	92,709	208,58	499,65	843,961	1,241,26	1,691,31
Cash flows at the End of Period	63,892	19,942	30,028	92,709	208,58	499,65	843,96	1,241,26	1,691,31	2,193,86

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 40: Expected Balance Sheet

Description	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
Assets										
Current Assets										
Cash	63,892	19,942	30,028	92,709	208,580	499,655	843,961	1,241,262	1,691,314	2,193,861
Inventory	30,785	43,099	55,413	67,727	80,041	92,355	104,669	116,983	129,297	141,611
Accounts Receivable	0	0	0	0	0	0	0	0	0	0
Total current Assets	94,677	63,041	85,441	160,436	288,621	592,010	948,630	1,358,245	1,820,611	2,335,473
Non-Current Assets										
Fixed Assets (net)	571,576	534,208	496,524	458,515	420,174	381,495	342,470	303,091	263,349	223,238
Total Non-Current Assets	571,576	534,208	496,524	458,515	420,174	381,495	342,470	303,091	263,349	223,238
Total Assets	666,253	597,249	581,965	618,951	708,796	973,506	1,291,100	1,661,336	2,083,961	2,558,711
Liabilities	-	-	-	-	-					
Current Liabilities										

Description	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
Payables	25,036	31,312	37,606	43,918	50,249	56,599	62,970	69,363	75,778	82,216
Remaining amount of Loan	85,114	93,183	102,017	111,688	-0	-0	-0	-0	-0	0
Total current Liabilities	110,150	124,495	139,623	155,606	50,249	56,599	62,970	69,363	75,778	82,216
Non-Current Liabilities										
Long Terms Loans	306,888	213,705	111,688	0	0	0	0	0	0	0
Total Long Term Liabilities	306,888	213,705	111,688	0	0	0	0	0	0	0
Total Liabilities	417,038	338,200	251,311	155,606	50,249	56,599	62,970	69,363	75,778	82,216
Owners' Equity										
Shareholders Contributions	313,164	313,164	313,164	313,164	313,164	313,164	313,164	313,164	313,164	313,164
Statutory Reseve	0	983	8,144	21,413	40,933	66,769	97,892	134,276	175,897	222,728
Retained Profits	63,950-	55,099-	9,346	128,768	304,450	536,973	817,074	1,144,533	1,519,122	1,940,603
Total Equity	249,214	259,049	330,654	463,345	658,547	916,906	1,228,130	1,591,973	2,008,183	2,476,496
Total Liabilities and Equity	666,253	597,249	581,965	618,951	708,796	973,506	1,291,100	1,661,336	2,083,961	2,558,711

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans, therefore the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 41: Free Net Cash flows Table

Net Cash flows	قبل التشغيل	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	القيمة النهائية
Net Free Cash flows	-782,910	25,42	78,32	132,3	184,9	238,1	291,0	344,3	397,3	450,0	502,5	6,871,252
Discount Factor	1.00	0.80	0.72	0.64	0.58	0.52	0.46	0.42	0.37	0.33	0.30	0.30
Net Present Value for Cash	-782,910	20,41	56,34	85,31	106,8	123,2	134,9	143,0	147,8	150,0	150,1	2,053,332

Table below illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates a total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table 42: Payback period

Payback Period	Before	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
----------------	--------	------	------	------	------	------	------	------	------	------	------

Operating											
Net Free	-782,910	25,422	78,326	132,363	184,957	238,147	291,075	344,306	397,301	450,052	502,547
Project Value	782,910	757,488	679,162	546,799	361,843	123,696	167,379	511,685	908,986	1,359,038	1,861,586
Payback Period	5	1	1	1	1	1	0	0	0	0	0

Table 43: Financial Analysis Results

	Value or Ratio
Weighted Average Cost of Capital (WACC)	%11.61
Net Present Value for Cash flows	2,388,672
Payback Period	5
Internal Rate of Return	%31.98

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Table 44: Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC
Original Scenario	%31.98	5	11.61%
Revenues declined by 10%	%29.35	5	11.61%
Operating Expenses Increased by 10%	%21.72	7	11.61%

Conclusions and Recommendations

Based on the profitability analysis and considering that project internal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 2,388,672 Jordan Dinars considering that the project provides 16 Job Opportunities for the governorate residents.